

SYLLABUS :-

1. Name of the subject Transportation Engineering Laboratory. New Number. 2. L-T-P 0-0-3 3. Credits 2 4. Name of the Department Department of Civil Engineering 5. Status of the subject .a. Specify the Session, 2021-22 .Spring. .b. Please specify the level of the subject : UG .c. Whether the subject will be offered as core or elective : Core .d. The semester in which the subject will be offered: Autumn .e. Name of the programme in whose curricula this subject will be included : B. Tech 6. Prerequisites for the subject, if any NIL 7. Objectives The laboratory experiments and others assignments proposed to be covered in Transportation Engineering Laboratory supplement the theoretical aspects to be covered in Transportation engineering and pavement engineering subjects .both core subjects. and will include characterization of different pavement materials including soils, aggregates, bituminous binders and bituminous mixes and planning, design and evaluation of transportation facilities 8. List of Experiments and Assignments Laboratory Experiments 1 Shape Tests on Aggregates - .a. Combined Flakiness and Elongation Index and .b. Superpave Elongated and Flaky Particles Test and .c. aggregate gradation .Cc, Cu. and comparison with Fuller's gradation 2 Determination of .a. Penetration and .b. Softening Point values of Bituminous Binders 3 Determination of .a. Ductility and .b. Viscosity of Bituminous binders 4 .a. MERLIN roughness and Dynamic Cone Penetrometer experiments on selected section of road and .b. impact test on aggregates 5 .a. Stripping Test .moisture damage. test for aggregates .b. Determination of bulk and theoretical maximum densities of Marshall specimen of bituminous mix 6 .a. Determination of CBR value of a given soil .b. Packing characteristics of aggregates of different gradations Assignments of Traffic and Planning 7 Different algorithms and ML Techniques and their applications in solving traffic and transportation problems: network connectivity, minimum spanning tree, shortest path, network flow, clustering techniques, time series analysis, demand forecasting, analysis of ordinal data